IS511E Blockchain in Business
Mid-term assessment 1, group project

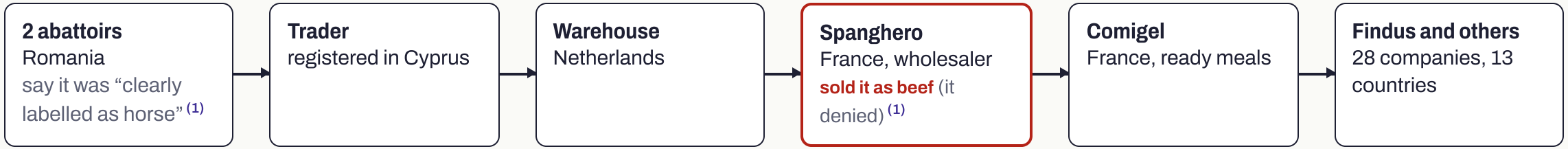
MeatTrace

A shared, tamper-evident record so that meat cannot change its label between suppliers

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Sector: food supply chain, processed meat in Europe
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In 2013, about 750 tonnes of horse meat were sold as beef across Europe, and nobody could quickly see where the label changed



What happened	Figure
Horse meat sold as beef, over about six months (French government) (1)	**≈ 750 t**
Ready-meal products made with it (French anti-fraud office) (1)	**≈ 4.5 million**
Companies and countries reached (1)	**28 in 13**
EU official tests on beef products positive for horse DNA (193 of 4,144) (2)	**4.66 %**

It is still happening

In a 2022 EU operation, **46 %** of 320 imported honey samples were found adulterated, some with forged traceability information. (3)

The law asks for traceability, link by link

EU Regulation 178/2002, art. 18: each firm must be able to identify its suppliers and its customers, the “one step back, one step forward” principle. (4) Nobody is required to see the whole chain.

(1) AFP via IOL, “Horsemeat: French company blamed”, 14/02/2013; Al Jazeera, “France says horse meat tip-offs ignored”, 15/02/2013.
(2) European Commission, horse meat test results, 2013.
(3) OLAF report 2023, “No money for fake honey”.
(4) EU Regulation 178/2002, article 18.

FreshMeals: our business case

The company we imagined

FreshMeals SA

A French ready-meal maker.

- 6 beef suppliers in 3 countries
- 1,500 beef batches a year
- €120 M annual sales

Business figures are our own assumptions.

The partners around FreshMeals

Abattoir
Prove correct labelling

→

Trader
Protect prices and suppliers

→

FreshMeals
Avoid recalls and brand damage

→

Retailers
Show proof of origin

Independent DNA lab
Tests batches and provides evidence

Food safety authority
Enforces the law and investigates

How it works today

- Companies exchange certificates and invoices by email.
- Each company keeps its own separate records.
- To trace a batch, FreshMeals must contact its suppliers and follow the chain backwards by phone.

The problem for FreshMeals

- Tracing a batch takes time.
- Documents can conflict or be changed.

Our goal

To trace each batch back to its abattoir in under 1 hour, with 100% of handovers signed.

FreshMeals and its business figures were imagined by our team. Stakeholders reflect the 2013 supply chain.

MeatTrace: every handover recorded on one shared history

How it works, for each batch

The sender records the batch number, species and weight, then signs digitally.

→

The receiver checks delivery and confirms it. If the declarations do not match, the system raises an alert.

→

For selected batches, an independent lab tests the meat and adds a signed result linked to the batch.

→

Authorized users can then follow the recorded journey back to the abattoir.

Value proposition

- Faster tracing across partner records
- Inconsistent declarations flagged for review
- More targeted recalls when history is complete

Benefits for partners

- Retailers: proof of origin for customers and inspectors
- Suppliers: evidence of what they did if a problem occurs

The record helps check the documents. DNA tests help check the meat.

Proposed workflow and expected benefits. Access depends on user roles. Pilot results remain to be measured.

Business model and KPIs

Funding model: consortium contributions

- FreshMeals, 2 retailers and the DNA lab share all costs equally: about **€25k per member per year**
- Suppliers and the authority pay nothing, which removes the main barrier to joining
- New buyers could join later and help share the network's costs.

Adoption path

Year 1
FreshMeals, 3 suppliers, DNA lab (pilot)

→

Year 2
All 6 suppliers and 2 retailers

→

Year 3
Other buyers of the same suppliers

Key performance indicators

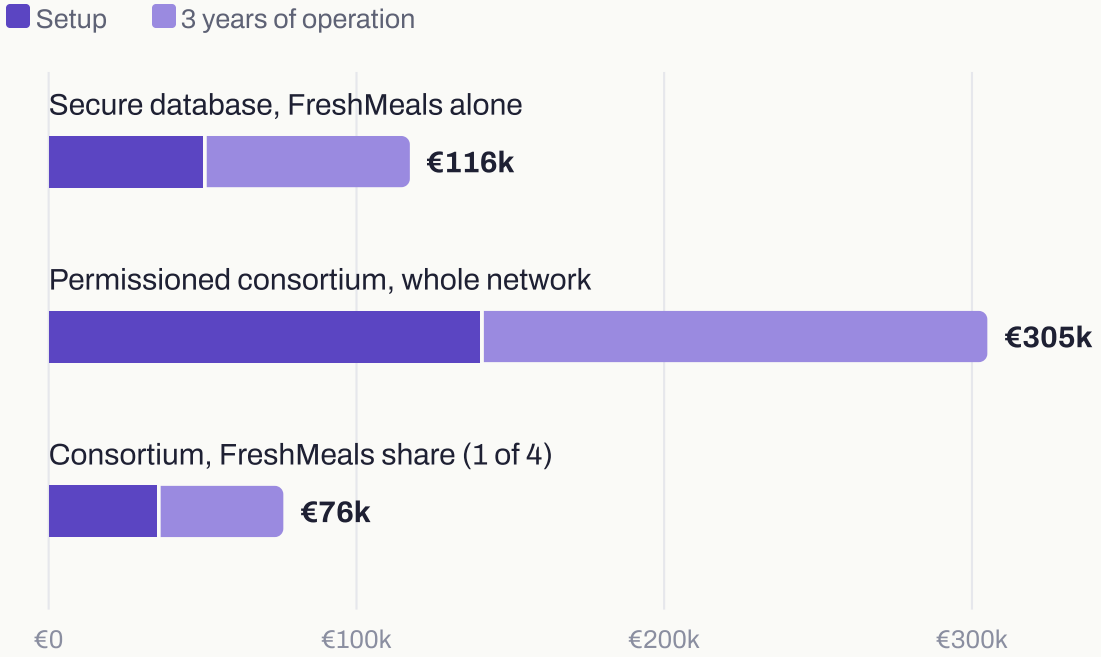
KPI	Today	Target after pilot
Trace-back time meal to abattoir, per batch	Several days	**< 1 hour**
Signed history beef batches with every handover signed	0 %	**100 %**
DNA coverage batches with a linked, signed DNA test	Rare, not linked	**≥ 10 %** random sample

Expected benefits

- A recall is limited to the guilty batches, not the whole product line. One full recall ≈ €2 M
- Trace requests answered in minutes instead of days of staff time
- Proof of origin sells: Carrefour said its blockchain-tracked chicken and pomelos sold better (no figures published) (1)

(1) ESM Magazine, "Carrefour says blockchain tracking boosting sales of some products", 2019.
Resources: development partner, hosting of 4 nodes, security, training, support, governance board. Figures are our own estimates.

The consortium costs more, but FreshMeals pays less when four members share it



3-year cost = setup + 3 × annual operation. All figures in euros, team estimates (details in appendix B).

Architecture	Setup	Annual	3 years
Secure database	€50k	€22k	**€116k**
Consortium, network	€140k	€55k	**€305k**
Consortium, 1 share	€35k	€13.75k	**€76k**

Main assumptions

- Developer day at €500. Database ≈ 80 days, ledger and portal ≈ 180 days
- Hosting: €6k a year for the database, 4 nodes × €5k for the network
- Setup includes security audit, onboarding and the legal charter
- Permissioned network: no cryptocurrency or transaction fees

Most uncertain assumption: the number of paying members. If FreshMeals had to pay alone (€305k), the secure database would win.

Extra cost of the network over a database: €189k in 3 years, less than 10 % of one full recall (≈ €2 M).

We choose a permissioned consortium, because several independent firms need one history and none of them can be the referee

Option	Trust	Privacy	Performance	Cost, 3 years	Governance	Verdict
Public blockchain	Anyone can verify	Batches and partners visible to the world	Slower, fees vary	Fees per transaction	No say for members	Rejected
Private blockchain run by FreshMeals	FreshMeals is judge and party	Good	Fast	Close to the consortium	One owner	Rejected
Permissioned consortium	Every member keeps a copy; nobody rewrites alone	Role rights; prices stay off-chain	Seconds per block	€305k, shared by 4	Board of members	**Chosen**
Secure database run by FreshMeals	Partners must trust FreshMeals' admins, who can edit history	Good	Fastest	€116k, cheapest	Simple	Fallback

When a database is enough

One trusted party runs the whole process, like a bank with its own accounts. A secure database is cheaper, faster and simpler, and we would recommend it.

Why it is not enough here

6+ independent firms in 3+ countries, competitors among them, blamed each other in 2013. Nobody accepts that a rival controls the history, and each needs proof it can show to the others.

Concepts from sessions 1 and 5: shared ledger, public, private and permissioned networks, cost and governance.

Each handover becomes a signed transaction, and blocks are chained by their hashes

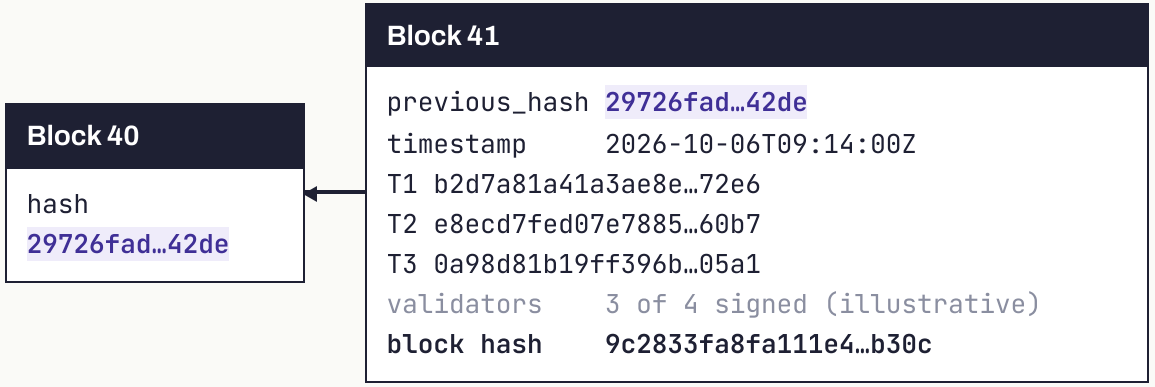
Transaction	Fields hashed	Signed by
T1 Slaughter	`batch_id` `species` `weight_kg` `site` `date`	Abattoir
T2 Transfer	`batch_id` `from` `to` `species_declared` `weight_kg`	Sender, then receiver confirms
T3 DNA test	`batch_id` `lab` `result` `report_sha256`	Independent lab

Hash. SHA-256 turns the fields into a 64-character fingerprint. Change one letter and the fingerprint changes completely.

Signature. The sender signs the fingerprint with its private key. Anyone checks it with the public key. It proves *who* issued the record and that it was *not changed*.

Off-chain. Prices, contracts and full lab reports stay in each firm's systems. Only their hash goes on the ledger.

Workflow. Create (abattoir records the batch), authorize (role check), verify (hash and signature), agree (3 of 4 validators), audit (shared history).



Hashes are real values computed by our Python prototype (slide 10); the validator line is illustrative.

If a partner signs something false, the hash cannot tell. A signature proves who said it, not that it is true. Random DNA tests catch false species, and the signer is liable under the consortium contract: penalty, then exclusion.

Concepts from sessions 2 and 3: transactions, blocks, previous-hash link, SHA-256, digital signatures.

Four known validators agree on each block, and roles decide who may create, approve or read

Role	Create	Approve	Read	Validate blocks
Abattoir	Slaughter records	No	Own batches	No
Trader	Transfer records	No	Batches it held	No
FreshMeals quality manager	Reception records	Yes, never own records	Full history of batches it buys	Yes
Independent DNA lab	Test results	Yes, batches it tested	Batches it tests	Yes
Retailer	No	No	Origin and tests of what it sells	Yes
Food safety authority	No	No	All records, for audits	Observer, no vote

Separate duties: nobody approves their own record, the approver is never the submitter, and no single member can accept a block alone (3 of 4 votes).

Consensus: 3 of 4 votes

Our 4 validators are FreshMeals, Retailer A, Retailer B and the DNA lab (the food authority only observes). They take turns proposing the next block. A block is accepted, and its place in the order fixed, when 3 of 4 sign it (BFT voting, like PBFT), so one dishonest member cannot fake or block it. We know all the members, so no mining (PoW) or staking (PoS) is needed.

- **Identity:** company certificate, plus named staff accounts with two-factor login
- **Authorization:** every request is checked against the role matrix
- **Separation of duties:** nobody approves a record they created
- **Confidentiality:** prices, contracts and full reports stay off-chain
- **Revocation:** leavers are disabled the same day; the board can revoke a company
- **Governance:** founders' board, one vote each. New members and rule changes need 3 of 4. The DNA result settles species disputes

Concepts from sessions 4 and 5: consensus, access rights, separation of duties, revocation, governance.

Python test A: changing horse to beef gives a completely different fingerprint

```
def record_hash(record):
    text = json.dumps(record, sort_keys=True)          # stable text
    return hashlib.sha256(text.encode()).hexdigest()   # SHA-256 fingerprint
```

```
relabelled_record = {**slaughter_record, "species": "BEEF"}   # new dict, original unchanged
relabelled_hash = record_hash(relabelled_record)

print("Original species  :", slaughter_record["species"])
print("Relabelled species:", relabelled_record["species"])
print("Approved hash  :", approved_hash)
print("Relabelled hash:", relabelled_hash)
print("Same record?   :", approved_hash == relabelled_hash)

Original species  : HORSE
Relabelled species: BEEF

Approved hash  : b2d7a81a41a3ae8e3847b21c821538803f7268b85e502cb86736d78f6d2272e6
Relabelled hash: b2c60303ead9a020cc82bbf4de7292c4885478ab67f2ec54c967d8992228bc52
Same record?   : False
```

```
    "species_declared": "BEEF",
}
signed_species = slaughter_record["species"]
status = "OK" if resale_transfer["species_declared"] == signed_species else "MISMATCH, batch blocked"

REC-004 declares       : BEEF
Signed history says    : HORSE
Check                  : MISMATCH, batch blocked
```

What it means for the business

- We changed one word (HORSE to BEEF) and the fingerprint is completely different.
- Every validator stores the approved fingerprint, so editing the abattoir's record, or an old block, no longer matches.
- In 2013 the fraud was a *new* document saying "beef". That is why we added a rule: the declared species must equal the signed history.
- **Limit:** the hash proves the record was not changed, not that the abattoir told the truth. That is the DNA test's job.

Screenshots of code and outputs from our Colab notebook IS511E_Midterm1_MeatTrace.ipynb (third block: the 2013 scenario, shortened).

Python test B: one function checks every request, with 2 actions allowed and 3 denied

```
role_permissions = {
    "abattoir":        {"create_record", "read_own"},
    "trader":          {"create_record", "read_own"},
    "quality_manager": {"create_record", "approve_record", "read_batch"},
    "lab_inspector":   {"create_test", "approve_record", "read_batch"},
    "auditor":         {"read_audit"},
}
```

```
def can_do(user, action, record_id=None):
    account = users[user]
    if not account["active"]:
        return False, "account revoked"
    if action not in role_permissions[account["role"]]:
        return False, f"role '{account['role']}' has no '{action}' right"
    if action == "approve_record" and records[record_id]["created_by"] == user:
        return False, "separation of duties: cannot approve own record"
    return True, "permission granted"
```

	Result	User	Role	Action	Record	Reason
0	ALLOWED	anna	lab_inspector	approve_record	REC-001	permission granted
1	ALLOWED	claire	quality_manager	read_batch	REC-001	permission granted
2	DENIED	pieter	trader	edit_record	REC-001	role 'trader' has no 'edit_record' right
3	DENIED	claire	quality_manager	approve_record	REC-007	separation of duties: cannot approve own record
4	DENIED	dan	trader	create_record	-	account revoked

Result	Who and what	Business risk handled
Allowed	anna, DNA lab, approves REC-001	An independent party checks the abattoir's record
Allowed	claire, FreshMeals, reads the batch history	The buyer checks origin before cooking
Denied	pieter, trader, edits REC-001 (no 'edit_record' right)	Stops relabelling horse as beef, the 2013 fraud
Denied	claire approves her own reception REC-007 (separation of duties)	No self-certification
Denied	dan, who left the trading firm, tries to create a record (account revoked)	An ex-employee cannot sign for a partner

Same 5 tests as the output on the left: 6 users, 5 roles, and no role has an `edit_record` right. In a real network this check would run on every node, not in a notebook.

Screenshots of code and outputs from our Colab notebook IS511E_Midterm1_MeatTrace.ipynb.

A 6-month pilot on one product line, then scale only if tracing takes under 1 hour

Proof of concept, Oct 2026
Done: hash test and permission rules in Python

→

Pilot, Jan to Jun 2027
FreshMeals, 3 suppliers, DNA lab, one product: beef lasagne. ≈ €70k of the €140k setup

→

Deployment, Jul 2027 to 2028
All 6 suppliers, 2 retailers, authority as observer; suppliers may run a node. Remaining ≈ €70k, then €55k a year

→

Improvement, from 2028
Quarterly KPI review, add chicken and pork, admit new buyers. €55k a year, shared

Risk	How we reduce it
A partner enters false data	Random DNA tests; receiver must confirm weight and species
Confidential data becomes visible	Role rights, least privilege, prices off-chain
Members disagree	Written charter, 3-of-4 vote, DNA result settles disputes
Costs exceed the plan	Pilot first; secure database as fallback
Partners do not join	Free for suppliers, simple portal. TradeLens (300+ members) announced its closure in 2022 for lack of industry collaboration [(1)]

Our recommendation

Build MeatTrace as a permissioned consortium of FreshMeals, 2 retailers and an independent DNA lab: **€305k over 3 years, €76k for FreshMeals.**

Value: batches traced in under one hour, relabelling caught at the next handover, fewer disputes.

Limits: it cannot see inside the meat, so DNA tests stay essential. Without 3 co-funders, choose the secure database.

First pilot target: 100 % of pilot batches traced to the abattoir in under one hour.

Still to build: real identities and certificates, network nodes, web portal, security audit.

(1) Container Magazine, “Maersk and IBM to discontinue TradeLens following lack of industry collaboration”, 30/11/2022. Costs per stage are our own estimates (appendix B).

Appendix A: references

1. Al Jazeera (15 Feb 2013). *France says horse meat tip-offs ignored.* aljazeera.com/news/2013/2/15/france-says-horse-meat-tip-offs-ignored
2. AFP, via IOL (14 Feb 2013). *Horsemeat: French company blamed.* iol.co.za/news/world/horsemeat-french-company-blamed-1470925. French government and anti-fraud office (DGCCRF) findings: ≈ 750 t over six months, ≈ 4.5 million products.
3. European Commission, DG SANTE (2013). *Horse meat 2013–14: test results.* food.ec.europa.eu/food-safety/eu-agri-food-fraud-network/eu-coordinated-actions/horse-meat-2013-14/test-results_en
4. European Anti-Fraud Office, OLAF (2023 report, 2022 operation “From the Hives”). *No money for fake honey.* ec.europa.eu/olaf-report/2023/investigative-activities/protecting-eu-revenue/no-money-for-fake-honey_en.html
5. Regulation (EC) No 178/2002, Article 18: traceability. eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32002R0178
6. LF Decentralized Trust (Hyperledger). *Case study: Walmart and Hyperledger Fabric* (mango trace: about 7 days to 2.2 seconds). lfdecentralizedtrust.org/case-studies/walmart-case-study
7. Container Magazine (30 Nov 2022). *Maersk and IBM to discontinue TradeLens following lack of industry collaboration.* container-mag.com/2022/11/30/maersk-and-ibm-to-discontinue-tradelens-following-lack-of-industry-collaboration
8. ESM Magazine (2019). *Carrefour says blockchain tracking boosting sales of some products.* esmmagazine.com/technology/carrefour-says-blockchain-tracking-boosting-sales-of-some-products-75860

Course material: IS511E sessions 1 to 5 (Pr. Imad El Harraki). FreshMeals, its suppliers, users and all business-plan figures were imagined by our team.

Appendix B: how we estimated the three-year costs (our estimates, in euros)

Secure database run by FreshMeals

Item	Setup	Annual
Web portal and database, ≈ 80 days × €500	40,000	
Security setup: two-factor login, encryption, penetration test	6,000	
Supplier onboarding and training	4,000	
Cloud hosting and backups		6,000
Maintenance and support, ≈ 24 days		12,000
Security monitoring and yearly audit		4,000
Total	**50,000**	**22,000**

3 years: 50,000 + 3 × 22,000 = **€116,000**

Permissioned consortium, 4 validator nodes

Item	Setup	Annual
Ledger, smart contracts and portal, ≈ 180 days × €500	90,000	
Network, identities and certificates	20,000	
External security audit	15,000	
Legal charter and onboarding	15,000	
Hosting, 4 nodes × €5,000		20,000
Maintenance and support, ≈ 50 days		25,000
Governance board, audits, certificate management		10,000
Total	**140,000**	**55,000**

3 years: 140,000 + 3 × 55,000 = **€305,000**; one of 4 shares = **€76,250**

Day rates and hosting prices are rough market estimates for a European IT service provider, to be confirmed by quotes during the pilot.